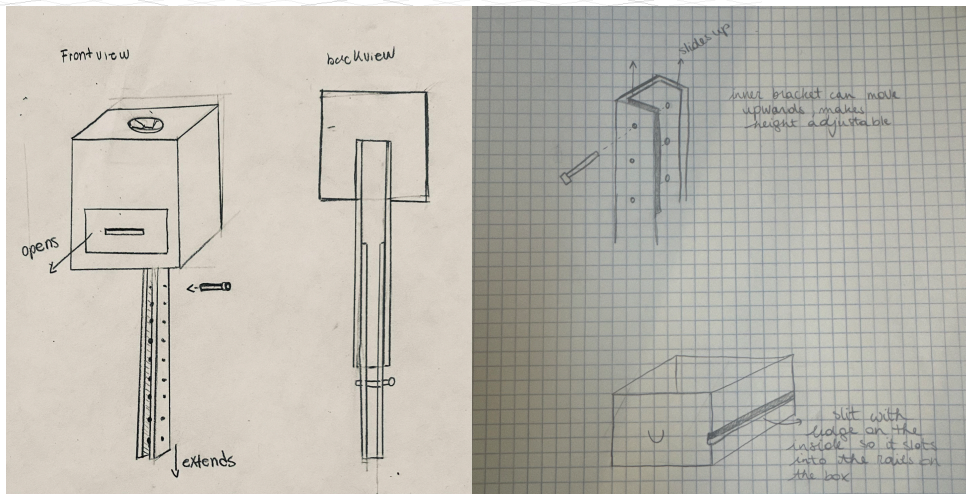
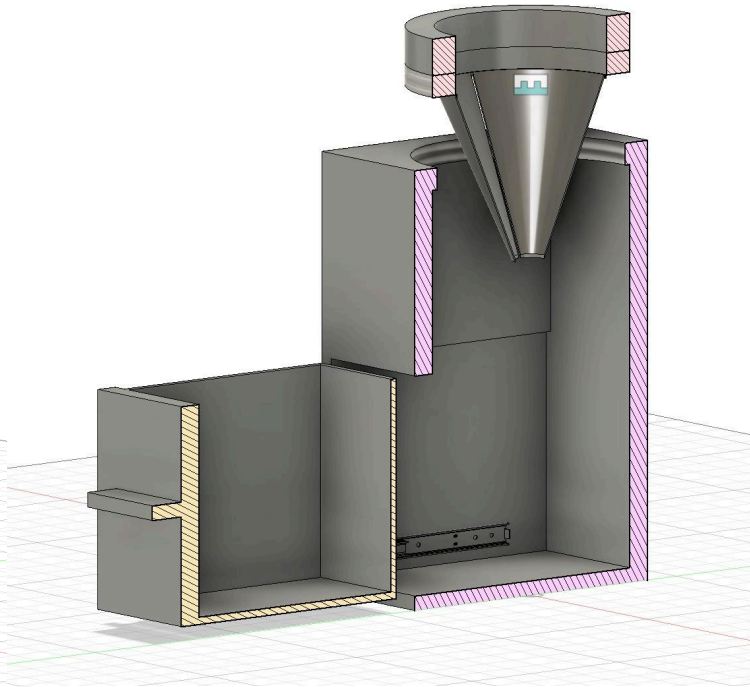
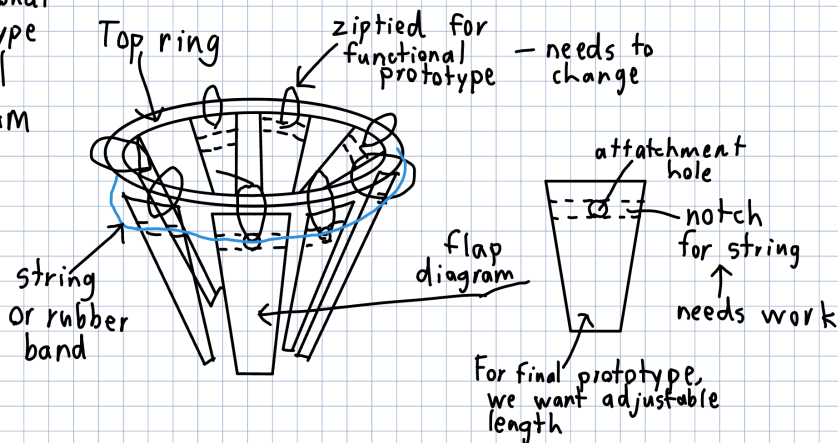


Design Documentation

- Container
 - Description: Outer housing of the trap which contains the drawer and trap funnel as well as interfaces with the extendable post
 - McMaster: N/A, all parts from Taylor design studio
 - Fabrication: Use bandsaw to cut wood to size, then glue sides together to form box
- Drawer
 - Description: a drawer on a set of rails at the bottom of the trap to catch the SLF and store the trap liquid
 - McMaster: none, materials include wooden sheets available in the Taylor design studio, CA glue
 - Fabrication: glue 2 strips of wood cut to size onto the inside of the box, make the box for the drawer out of wood and cut a slit with a ledge at the bottom of the slit on either side of the drawer such that it slides into the container easily
- Trap Funnel
 - Description: Gonna need a photo
 - McMaster: No McMaster parts for functional prototype. Materials: diameter 8 cm wire, wood, zipties, string, and rubber bands from TDS.
 - Fabrication: Used the bandsaw to cut flaps out of wood and used the bandsaw to cut a notch in the back of each for a rubber band/string to sit in. Drilled a hole in each flap with a hand drill. Used pliers to cut the wire and bent it into a circle. Ziptied each flap to the ring.
- Extendable Post
 - Description: two horseshoe shaped poles, where the smaller one fits into the bigger one, with holes drilled into the sides such that a rod can fit through respective holes and make the length adjustable
 - McMaster: no McMaster parts, materials: wooden sheets and cylinders available in Taylor design studio, CA glue,
 - Fabrication: sawed wood to desired width and length with the bandsaw, drilled holes (1cm diameter) every 2 inches in 2 out of 3 wooden parts of the same length, glued to strips of wood in a horseshoe shape, with the strips of wood with holes facing each other, for the inner and the outer part of the pole
 - Cut the wood cylinder (1cm diameter) into 4 inch long pieces, superglued ticker wooden cylinder at the end such that these pieces slot into the drilled holes on the pole.



Functional Prototype Funnel Diagram



ODP 5

1. Adult Dimension SLFs

- Length : 2.5 cm
 - Width (at rest) : 1.3 cm
 - Wingspan : 3.8 - 5 cm
 - Weight : 0.5 g
 - Height $\approx$ 2 cm
- } Average : $1.3 + 5 = 3.15$

$$\text{Volume} \approx 2.5 \cdot 3.15 \cdot 2 = 15.75 \text{ cm}^3$$

If the container fits $\sim$ 250 SLFs

$$250 \cdot 15.75 = 3937.5 \text{ cm}^3$$

$$\text{Container} : 20 \text{ cm} \times 20 \text{ cm} \times 30 \text{ cm} = 12,000 \text{ cm}^3$$

$$\text{Drawer} : 20 \text{ cm} \times 20 \text{ cm} \times 15 \text{ cm} = 6000 \text{ cm}^3$$

$$\text{Maximum Weight} : 20 \text{ cm} \times 20 \text{ cm} \times 15 \text{ cm} \times 1 \text{ g/cm}^3 \approx 12 \text{ kg}$$

McMaster

- Rails the drawer
 - 11435A202 \$23.14
 - <https://www.mcmaster.com/11435A202/>
- Adjustable zip ties:
 - 7134K148 \$9.38 for a pack of 100 <https://www.mcmaster.com/7134K148/>
- Extension post
 - 1471T51 6 feet \$21.58 <https://www.mcmaster.com/1471T51/>
 - 1471T53 4 feet \$20.30 <https://www.mcmaster.com/1471T53-1471T532/>
- Acrylic
 - 8505K721
 - <https://www.mcmaster.com/8505K721-8505K112/>

- \$9.65 each for 12 in by 12 in

Proposed McMaster Order:

Part Name	SKU	Price Per Unit	Units Desired
Spring Variety Pack	98596A790	\$11.44	1
Mini Drawer slides	3351N12	\$32.36	1
4 ft of aluminum U channel	9001K685	\$8.17	1
Aluminum L stock	8982K96	\$2.40	16
Aluminum 6061 Plates	89015K171	\$6.39	2
Pins to go through U channel	98381A047	\$3.33	2

- [8982K19](#) - L bracket

Design Test

- Container
 - Assembly and fastening: container is secure and stable on pole and can withstand 10 mph wind.
 - Test: Mount Pole and container, attach rope to the part of the container where the wind would typically hit. Pull horizontally until force is equivalent to the force applied by wind at 10mph. $F = \frac{1}{2} (\rho C A v^2)$, using pull scale.
 - What we will observe: Does the container stay in its original position relative to the pole (no rotation or loosening)? Does the pole bend? Is there any permanent deformation?
 - Success criteria: Measure the distance between the bottom end of the pole and the bottom end of the container, before and after force is applied. Distance should not change if the prototype is resistant enough.
- Drawer
 - Load-bearing performance: Can hold 12 kg without limitation of motion
 - Motion and interference: opening & closing smoothly without jamming
 - Repeating-Use: Drawer can be reused 50 times without visible damage
- Funnel
 - The flaps meet between 1 and 3 centimeters from each other in the middle of the funnel.
 - When a model of a similar size as a lanternfly is pulled on a string upward through the funnel, the flaps mesh together and prevent the model from escaping.
 - A string or rubber band that keeps the flaps pulled inward to the center of the funnel can be replaced as necessary to adjust the trap.

- The friction between the ring and flaps can be adjusted, and flaps can be replaced as necessary to allow easy entry of lanternflies into the trap.
- A model of the same size as an adult lanternfly moves through the funnel into the trap when dropped into the funnel.
- Extendable Pole
 - Motion and interference: Pole can move up and down smoothly without jamming
 - Repeating-Use: Position can be changed 50 times without visible damage (after 50 times lubrication reapplied) , pin can be pulled out with minimal force
 - Stability Support: Doesn't slip when changing height
 - Load-bearing: make sure the pole can hold maximum weight (14.25 kg)¹
 - Tests: Repeated use: Cycle the pole 50 times then inspect for physical damage; scratches, chipping, the deformation of the original horseshoe shape. Stability support: Mark initial height, apply load, measure any vertical displacement over time, check 20 minutes. :
 - Success criteria: There should be negligible visible changes to the pole and negligible vertical displacement when the realistic load is applied.

Results

- Container
 -
- Drawer
 - The drawer prototype was tested with loads up to 6 kg, which is approximately half of the target load for the final design due to anticipated differences in material strength. The results indicated that while the structure could support this load, the handle design requires improvement to reliably sustain higher forces during repeated pushing and pulling. Additionally, the drawer's motion was evaluated under a 3 kg load, where it successfully opened and closed smoothly for 15 consecutive cycles without jamming. Additionally, the drawer can be reused 50 times without visible damage. These results suggest that the sliding mechanism is effective, but further refinement of the handle is necessary to meet the full design requirements.
- Trap Funnel
 - Under good trap conditions, the fly model could be placed inward on a string and with very little force fall through the funnel, and when pulled back up by string the fly model could not escape once inside the trap
 - These so called good conditions were when the flaps were properly spaced and oriented, and the string tied around them was not caught
 - In worse conditions, such as improper spacing, twisted flaps, or a caught string, the trap either did not let the model bug enter, or could not keep it from escaping.

Implications:

- The flaps need spacers to keep them in place on the ring

¹ Weight max calculation:

- Weight of flies - 250 g
 - Fill container halfway full of sap (assume same density as water) - 12kg
 - weight of materials ~ 2 kg
- Total: 14.25 kg

- The flaps need more than one attachment point to keep the face of each flap pointing at the center of the ring
 - We need a better mechanism to prevent the flaps from opening too wide so that they will always be hit by a bug trying to escape - perhaps a ring near the bottom of the flaps keeping them close to the center
 - We need to make flaps that can be adjusted in length
 - The attachment of the funnel to the rest of the trap needs to not interfere with the action of the funnel and the range of motion of the flaps
 - Flaps need to be thinner so that their thickness does not cause them to collide with and get stuck behind adjacent flaps
 - Will consider moving from a circular funnel to a different shape like an octagon so flaps fit together better and do not have to deal with flaps on a curved hinge
- Extendable Pole
 - Pole position successfully raised up and down 50 times without visible damage or slippage
 - Pole position setting pin can be pulled with minimal force
 - Pole did not visibly deform when trap loaded with 6 kg
 - These show the extendable pole mechanism passed all tests we set for it and needs minimal change for our next iteration

Success Criteria

Our project is a bug trap for spotted lantern flies to minimize the harm they can do on crops of a vineyard. Our trap is box shaped and will be placed on an adjustable pole in the vineyard, it will contain a cone shaped trapping mechanism and a drawer to catch the SLF's and remove them.

- Drawer weight capacity, this is assessing if the drawer could successfully hold the weight of the maximum load SLF in addition to a safety factor, success is determined if the drawer can easily be closed 20 times with a load of 12kg and without undergoing any visible damage
 - Can be demoed on exhibit day
- Drawer can be slid on and off the rails easily, it is assessing the practical accessibility of the drawer, success is determined if the drawer is opened and closed 50 times without getting stuck
 - Can be demoed on exhibit day
- Moving the pole up and down, it is assessing the durability of the pole mechanism and the practicality of the mechanism, success is determined if it can be adjusted 50 times without damaging the mechanism.
 - Can be demoed on exhibit day
- Removability and adjustability of the length of the flaps, it is assessing flaps are easily adjustable depending on circumstances and testing with real bugs, success is determined if the flaps can be successfully removed and reattached and their length can be adjusted (50 times each) without damaging them.
 - Can be demoed on exhibit day
- The flaps range of motion is constrained so that they stay close to the center of the funnel: they cannot move past the vertical. This is to keep the opening at the bottom of the funnel small so that

anytime a fly tries to leave the trap, they will hit the flaps. The funnel opening goes back to its original position (original diameter opening) after our demo fly goes through it (30 times). This is assessing the range of motion of the flaps, and success will be determined if the user can set how far they'd like the flaps to be able to move and the position they are in as a default.